

Telecom Complaints Mining — Hybrid Clustering Report

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1 Objective

Analyze telecom complaints and extract actionable patterns to support decision-making. Workflow: **EDA** → **text preprocessing** → **hybrid features** → **clustering** → **insights + prioritization**.

2 Dataset

Dataset contains telecom complaints with fields: `wilaya`, `canal`, `texte_reclamation`, `categorie`, `sentiment`, `urgence`, `status`. We keep `categorie` for reporting but do not treat it as ground truth for training due to observed inconsistency with text.

3 What we implemented (Notebook steps)

1. **Load + clean:** parse dates, strip categorical values, create time features (month/weekday).
2. **EDA:** distributions by wilaya/channel/urgency/status/sentiment and complaints-per-month trend.
3. **NLP preprocessing:** lowercase, remove punctuation, remove stopwords, keep informative tokens.
4. **Text features (TF-IDF):** represent each complaint text using uni/bi-grams TF-IDF.
5. **Metadata features (One-Hot):** encode wilaya/channel/urgency/sentiment/status.
6. **Hybrid representation:** normalize each block and concatenate $X_{hybrid} = [\alpha \cdot \hat{X}_{text} \mid \beta \cdot \hat{X}_{meta}]$.
7. **Clustering:** run **KMeans** on X_{hybrid} and select k by **silhouette score**.
8. **Interpretation:** label clusters using top keywords + examples + operational profile.
9. **Prioritization:** compute a priority score using volume + high urgency + unresolved + negative sentiment.

4 Algorithms used (brief + why)

- **TF-IDF:** transforms text into weighted keyword vectors to capture complaint themes.
- **One-Hot Encoding:** transforms categorical metadata into numeric features.
- **Hybrid features:** combines theme (text) + operations context (metadata) for actionable clusters.
- **KMeans:** stable, fast segmentation; easy to report at scale.
- **Silhouette score:** chooses the best number of clusters without ground-truth labels.

5 Results: cluster labels and characteristics

Final model produced 10 clusters. Table 1 lists labels and key rates.

ID	N	Label (theme)	High urg.	Escalated
0	10017	Installation delay (installation non réalisée / délais)	19.78%	5.04%
1	10048	Line noise issue (bruit sur ligne téléphonique)	20.28%	4.73%
2	10066	Incorrect billing / invoice correction (facturation incorrecte)	19.53%	4.99%
3	9909	Login/authentication error (erreur d'authentification / connexion)	20.26%	5.58%
4	9933	Offer/contract information request (demande d'information)	20.52%	5.10%
5	9989	Service outage since morning (coupure de service)	20.20%	5.11%
6	10034	Low internet speed at night (débit très faible en soirée)	19.88%	5.06%
7	9896	Modem sync failure (modem ne se synchronise pas)	19.54%	5.01%
8	10112	TV freezing on some channels (image figée sur certaines chaînes)	20.32%	4.78%
9	9996	Cancellation issue + continued billing (résiliation)	19.93%	5.33%

Table 1: Hybrid KMeans clusters: size, label, high urgency rate, and escalation rate.

6 Priority matrix (what to fix first)

We rank clusters using: **volume + high urgency + unresolved + negative sentiment**. Top priorities are shown in Table 2.

ID	N	Theme	High urg.	Unresolved	Neg. sent.	Score
8	10112	TV freezing on some channels	20.32%	59.70%	60.62%	0.6611
1	10048	Line noise issue	20.28%	60.03%	60.06%	0.6583
0	10017	Installation delay	19.78%	60.16%	61.45%	0.6582
2	10066	Incorrect billing / invoice correction	19.53%	60.27%	59.54%	0.6569
9	9996	Cancellation issue + continued billing	19.93%	59.99%	60.41%	0.6558

Table 2: Top priority clusters using volume + urgency + unresolved + negative sentiment.

7 Direct insights from clustering

- **Recurring themes:** the model discovered 10 stable complaint themes (installation delays, billing issues, outages, low speed, TV freezing, etc.).
- **Most negative theme:** Installation delay (cluster 0) has the highest negative sentiment (61.45%).
- **Top priority theme:** TV freezing (cluster 8) ranks first by priority score (0.6611) with high unresolved (59.70%) and negative sentiment (60.62%).
- **Main escalation-risk theme:** Authentication/login issues (cluster 3) have the highest escalation rate (5.58%).

8 Impact on decision-making

- **Prioritization:** start improvement actions with clusters 8, 1, 0, 2, 9 because they rank highest by the priority score.

- **Routing/escalation:** treat cluster 3 as the main escalation-risk theme; route these tickets directly to specialized support.
- **Customer experience focus:** improve installation scheduling/communication for cluster 0 because it drives the highest negative sentiment.

9 How expected outcomes were achieved

- **Frequent complaint types + distribution across regions/channels:** clusters provide complaint types; profiles show top wilayas and channels per type.
- **Urgency and resolution patterns:** cluster-level high-urgency, unresolved, and escalation rates quantify operational risk by theme.
- **Prioritization + recommendations:** the priority matrix ranks themes to address first and supports routing and staffing decisions.

10 Limitations

Texts are highly templated and category labels may be inconsistent with the text. Therefore, conclusions are strongest for clustering-based segmentation and prioritization rather than supervised prediction.